

Reproduction in Humans P2 questions

Compiled by: [**Mustafa Asif**](#)

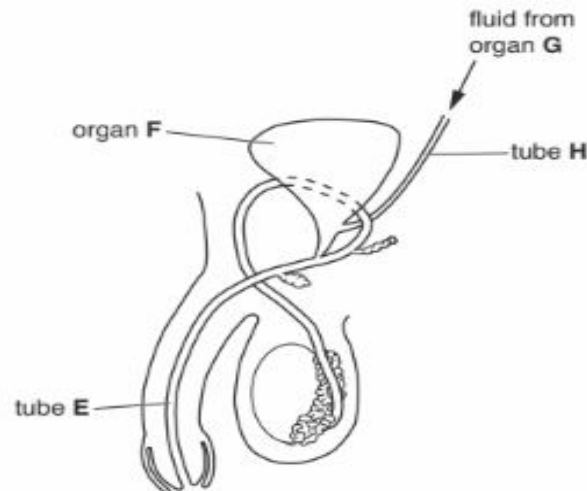
- (q) identify on diagrams of the male reproductive system and state the functions of the testes, scrotum, sperm ducts, prostate gland, urethra and penis
- (r) identify on diagrams of the female reproductive system and state the functions of the ovaries, oviducts, uterus, cervix and vagina
- (s) compare male and female gametes in terms of size, numbers and mobility
- (t) describe the menstrual cycle, with reference to the alternation of menstruation and ovulation, the natural variation in its length and the fertile and infertile phases of the cycle
- (u) explain the role of hormones in controlling the menstrual cycle (including FSH, LH, progesterone and oestrogen)
- (v) describe fertilisation and early development of the zygote simply in terms of the formation of a ball of cells that becomes implanted in the wall of the uterus
- (w) state the function of the amniotic sac and the amniotic fluid
- (x) describe the function of the placenta and umbilical cord in relation to exchange of dissolved nutrients, gases and excretory products (**no** structural details are required)
- (y) describe the special dietary needs of pregnant women
- (z) describe the advantages of breast milk compared with bottle milk
- (aa) describe the following methods of birth control:
natural, chemical (spermicides), mechanical, hormonal and surgical
- (bb) explain that syphilis is caused by a bacterium that is transmitted during sexual intercourse
- (cc) describe the symptoms, signs, effects and treatment of syphilis
- (dd) discuss the spread of human immunodeficiency virus (HIV) and methods by which it may be controlled

Reproduction in Humans P2 questions

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0/N18/22/Q2

The diagram shows the human male reproductive organs and associated structures.



(a) (i) Identify each of the following:

tube E

organ F

organ G

tube H

[4]

(ii) State **one** difference between the fluids carried by tube E and tube H.

.....
.....[1]

(b) (i) State **one** way in which the fluid from organ G may be different in a person with diabetes compared to a person without diabetes.

.....
.....[1]

(ii) A person with diabetes may be treated with insulin produced by genetically modified bacteria.

Outline how such genetically modified bacteria may be produced and used to manufacture human insulin on a commercial scale.

.....
.....
.....
.....
.....
.....[4]

[Total: 10]

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(a) Describe the role of each of the following hormones in the menstrual cycle:

.....

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 104

.....

[illegible]

- =====

.....

.....

.....[2]

- name of method

.....

[illegible]

.....

.....[1]

[1]

FACEBOOK.COM/CAMBRIDGE O LEVEL/IGCSE (GROUP)

Reproduction in Humans P2 questions

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M/J18/21/Q2

The diagram shows a male human gamete.



- (a) (i) State the name of the male human gamete.

.....

[1]

The diagram below shows the male reproductive system and associated organs.



- (ii) Label with a letter **X** on the diagram where the male gametes are produced.

[1]

- (iii) The nucleus of the male gamete is different from the nuclei of other types of cell found at location **X**.

State the cause of this difference and explain its importance in reproduction.

cause of difference

.....

.....

explanation

.....

.....

.....

[3]

- (b) Describe the differences in size and mobility between the male human gamete and the female human gamete.

.....

.....

.....[2]

- (c) Complete the paragraph by writing the most appropriate word in each of the spaces.

The fusion of a male human gamete and a female human gamete to form a is called A ball of cells is then formed that becomes implanted in the wall of the

[3]

[Total: 10]

Reproduction in Humans P2 questions

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M/J18/21/Q5

- (a) (i) Describe the main characteristics of a virus.

.....

.....

.....

.....[3]

- (ii) The human immunodeficiency virus (HIV) reproduces inside white blood cells and destroys them.

Use your knowledge of the functions of white blood cells to suggest why the virus is named the immunodeficiency virus.

.....

.....

.....[2]

- (b) (i) HIV causes a disease called AIDS. The virus may be transmitted during sexual intercourse.

State two methods by which the spread of HIV by sexual intercourse may be controlled.

1

.....

2

.....[2]

- (ii) HIV may be transmitted in other ways.

State two ways, other than during sexual intercourse, by which HIV may be transmitted.

1

.....

2

.....[2]

[Total: 9]

Reproduction in Humans P2 questions

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0/N17/22/Q1

Fig. 1.1 shows stages in the development of human twins.

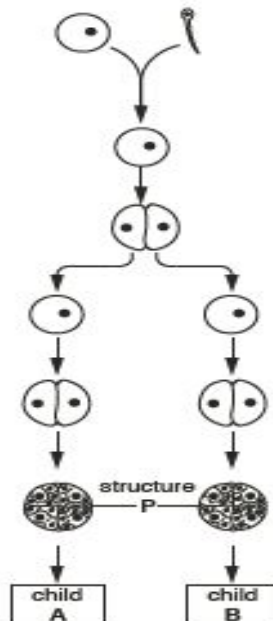


Fig. 1.1

(a) On Fig. 1.1, label and name each of the following:

- a gamete,
- a zygote.

[2]

(b) Name the part of the female reproductive system that structure P enters.

.....[1]

(c) If the sex chromosome in the sperm is a Y chromosome, and in the ovum (egg) is an X chromosome, state the sex of child A and of child B. Explain your answer.

child A

child B

explanation

.....

.....

.....

.....

[4]

(d) Explain how a woman's body prevents further ova (eggs) from being released until the end of her pregnancy.

.....

.....

.....

.....

.....

[4]

[Total: 11]

Reproduction in Humans P2 questions

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0/N16/21/Q4(b)

- (b) (i) Many mothers choose to feed their child breast milk rather than bottle milk.

Describe some advantages of breast milk compared with bottle milk.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (ii) Suggest **one** reason why a mother may choose to feed her child bottle milk.

.....

.....

..... [1]

[Total: 11]

Reproduction in Humans P2 questions

Compiled by: [Mustafa Asif](#)

0/N16/21/Q6

Syphilis is an infectious disease.

(a) Describe how syphilis is transmitted.

.....

.....

.....

.....

.....

..... [3]

(b) Describe the symptoms, signs and effects of syphilis.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(c) Describe the treatment of syphilis and suggest why it is **not** possible to treat the infection that leads to AIDS in the same way.

.....

.....

.....

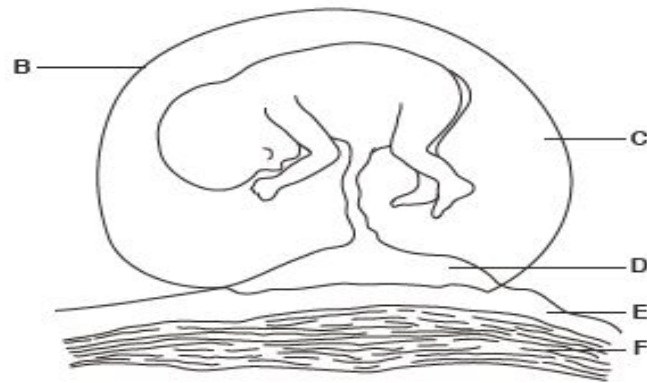
.....

..... [3]

[Total: 10]

Compiled by: Mustafa Asif

2 Fig. 2.1 shows a fetus developing inside its mother.



(a) Complete Table 2.1, using letters from Fig. 2.1, to identify each of the following:

	letter
a structure that contains urea released by the fetus	
a region that contains cells, almost all of which could be used for determining the sex of the fetus	
a structure that is used to expel the fetus at birth	
a temporary structure that would have been expelled during menstruation if pregnancy had not occurred	

(b) Every day, babies are born suffering from severe withdrawal symptoms as a result of their mothers taking drugs during pregnancy.

- ***** [1]

- [4]

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Reproduction in Humans P2 questions

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0/N15/21/Q4

Fig. 4.1 shows changes in the thickness of the uterus lining and in the concentration of progesterone in the blood during the menstrual cycle.

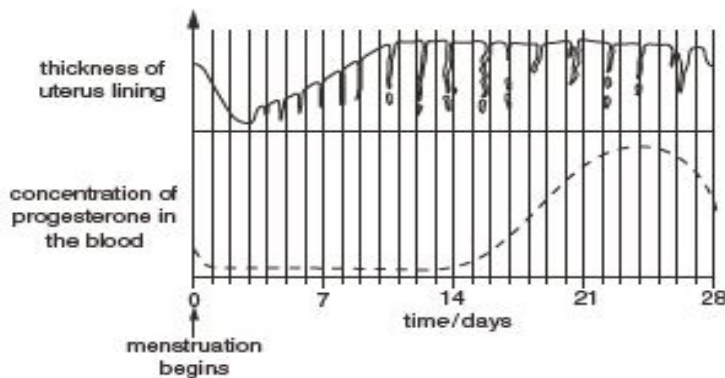


Fig. 4.1

- (I) Use Fig. 4.1 to state the time at which each of the following occurs:

the uterus lining reaches maximum thickness

.....

the concentration of progesterone begins to increase.

.....

[2]

- (II) Draw a ring around the day from Fig. 4.1 on which ovulation is most likely to occur.

day 0

day 7

day 14

day 21

day 28

[1]

- (III) Explain how it can be concluded from Fig. 4.1 that the person did **not** become pregnant during the 28 days shown, even though ovulation occurred.

.....

 [2]

- (b) The menstrual cycle is controlled by hormones. Name **two** hormones, **other than progesterone**, that control the menstrual cycle. State **one** role of each hormone.

name of hormone

.....

role in the menstrual cycle

.....

name of hormone

.....

role in the menstrual cycle

.....

..... [4]

[Total: 9]

Reproduction in Humans P2 questions

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M/J15/21/Q3

Fig. 3.1 shows a fetus developing in the uterus of a mother. The fluid labelled C contains cells from the fetus.

A long, hollow needle may be used to withdraw some of the fluid into a syringe. The DNA from the cells in this fluid can then be analysed to find the sex of the fetus and to detect mutations.

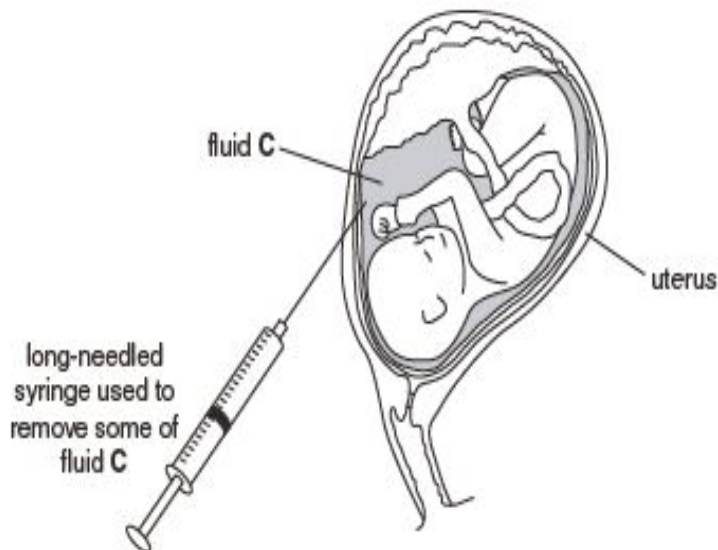


Fig. 3.1

- (a) Name fluid C and state its function.

name of fluid

function

[2]

- (b) (i) Label the placenta on Fig. 3.1 using a line and the letter P.

[1]

- (ii) State **two** functions of the placenta.

1

.....

2

.....

[2]

Reproduction in Humans P2 questions

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- (c) Fig. 3.2 shows the chromosomes found in the nucleus of one cell of a developing fetus.

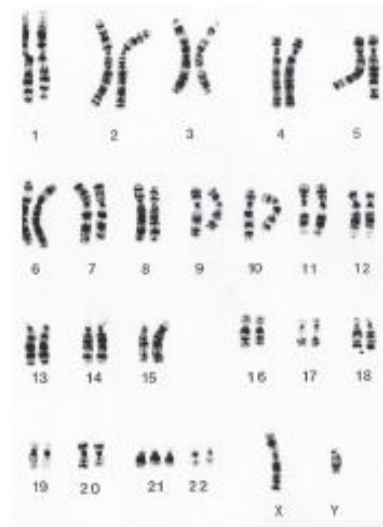


Fig. 3.2

State the sex of this fetus and explain your answer.

sex

explanation

[2]

- (d) This fetus has a mutation.

(i) Describe the mutation shown in Fig. 3.2.

.....
.....
.....
.....

[2]

(ii) Suggest the condition that this child could be born with as a result of this mutation.

.....

[1]

[Total: 10]

Reproduction in Humans P2 questions

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M/J15/22/Q9

Syphilis and HIV are both spread by sexual contact.

(a) Describe the cause and symptoms of syphilis.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

(b) State how syphilis is treated.

.....

..... [1]

(c) Discuss how the spread of HIV may be controlled.

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

Reproduction in Humans P2 questions

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0/N14/21/Q3

- (a) At puberty, hormones are produced that are responsible for the development of secondary sexual characteristics.

(i) Name **one** secondary sexual characteristic in males.

..... [1]

(ii) Name the hormone responsible for the development of secondary sexual characteristics in males.

..... [1]

(iii) Name the organs that produce this hormone. [1]

- (b) Fig. 3.1 shows the concentrations in the blood of two hormones, F and G, involved in a woman's menstrual cycle.

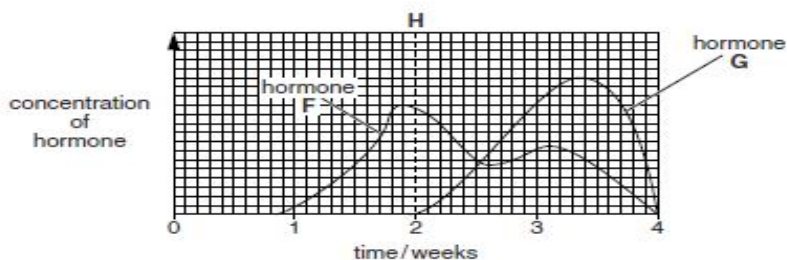


Fig. 3.1

(i) Name the hormones F and G.

F

G [2]

(ii) State what occurs at time H. [1]

- (c) Fig. 3.2 shows the thickness of the same woman's uterus lining over a 4-week period.

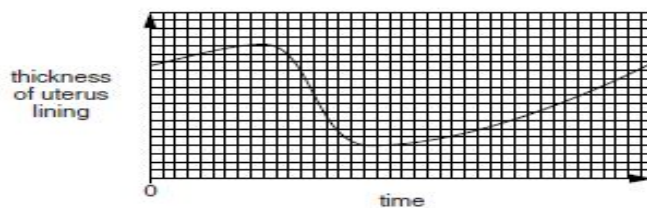


Fig. 3.2

The graph shown in Fig. 3.1 does not begin at the same time as the cycle shown in Fig. 3.2.

Indicate on Fig. 3.2, with a line labelled J, the stage shown by line H on Fig. 3.1, and explain your reason for choosing this point on the graph.

.....

 [4]

[Total: 10]

Reproduction in Humans P2 questions

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M/J14/22/Q4

- 4 Fig. 4.1(a) shows the reproductive organs of a man and Fig. 4.1(b) shows the reproductive organs of a woman.

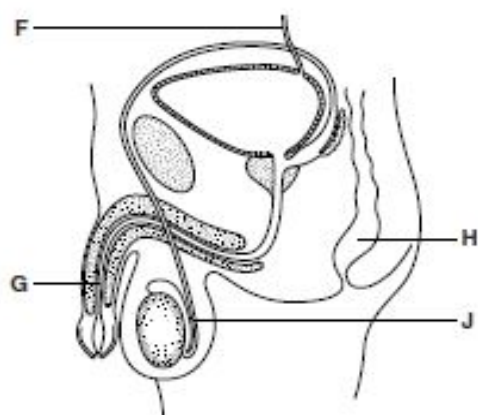


Fig 4.1(a)



Fig. 4.1(b)

- (a) Complete Table 4.1, stating the names of the structures in Fig. 4.1(a) and indicating whether they carry urine and/or sperms using yes **or** no as appropriate.

Table 4.1

structure identified by letter	name of structure	carries urine (yes or no)	carries sperms (yes or no)
F			
G			
H			
J			

[4]

- (b) Indicate by drawing a line across each of the relevant parts **In both Fig 4.1(a) and 4.1(b)**, where a cut may be made in order to carry out a form of surgical contraception.

[2]

- (c) In older men, the prostate gland tends to increase in size. Suggest an explanation for how this may affect urination.

.....

.....

.....

.....[2]

[Total : 8]

Reproduction in Humans P2 questions

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0/N13/22/Q9(a)

- (a) Describe the methods by which the spread of human immunodeficiency virus (HIV) may be controlled.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

Reproduction in Humans P2 questions

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M/J13/21/Q6

- (a) The menstrual cycle is controlled by several hormones including FSH and progesterone. Use your knowledge of the role of these hormones in the menstrual cycle to suggest explanations for each of the following.

- (i) FSH is given during fertility treatment to women who experience problems becoming pregnant.

.....

.....

.....

.....

.....

..... [3]

- (ii) Progesterone is a component of the contraceptive pill.

.....

.....

.....

.....

.....

..... [2]

- (b) With reference to **named** substances, describe the functions of the placenta and the umbilical cord.

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 10]

Reproduction in Humans P2 questions

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M/J12/21/Q7

- 7 (a) (I) State **one similarity** and **one difference** in the functions of the urethra in a male and in a female adult person.

similarity

.....

difference

..... [2]

- (II) State the **differences** between male and female human gametes in terms of size, numbers and mobility.

.....

.....

.....

..... [3]

- (b) Describe the **advantages** and **disadvantages** of surgical methods of birth control.

advantages

.....

.....

.....

.....

disadvantages

.....

.....

.....

.....

.....

..... [5]

[Total: 10]

Reproduction in Humans P2 questions

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0/N11/21/Q7

- 7 (a) Describe how a fully-formed fetus in the uterus develops from the time of fertilisation.

This image shows a full page of primary-ruled paper. It features multiple sets of horizontal dashed lines spaced evenly down the page, providing a guide for handwriting practice. In the bottom right corner, there is a small blue square icon containing a white number '7'. The rest of the page is blank white space.

- (b)** Describe the disadvantages of feeding a baby with bottle milk.

(This area contains faint bleed-through from the reverse side of the page.)

[Total: 10]

Reproduction in Humans P2 questions

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M/J11/21/Q3

Fig. 3.1 shows some stages in human reproduction.

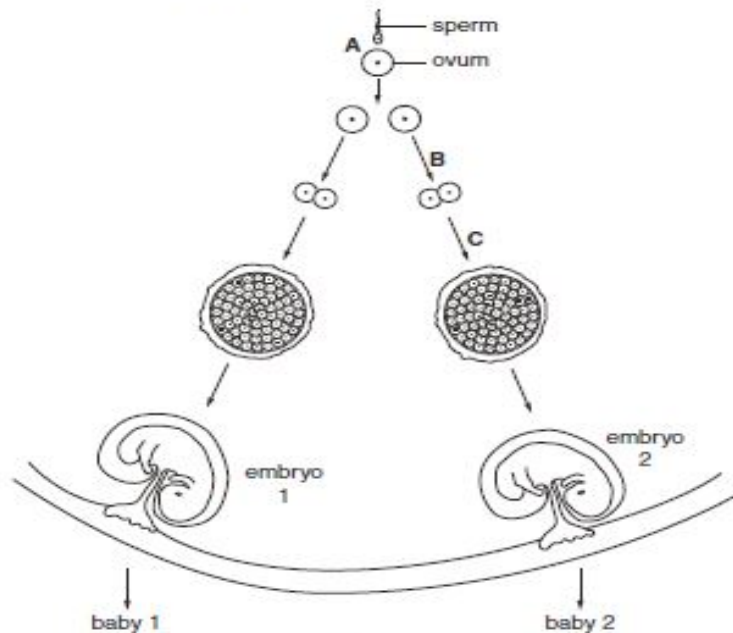


Fig. 3.1

- (a) Name the process about to occur at **A** and state where in the reproductive system it normally occurs.

process

where it occurs [2]

- (b) Name the type of cell division taking place at **B** and **C**.

..... [1]

The two embryos develop, are born and grow to become adults.

- (c) Suggest two ways in which these two adults must be **similar** to each other, and explain your answers.

1.

2.

explanation

.....

..... [4]

- (d) Suggest two ways in which they may **differ** from one another, and explain your answers.

1.

2.

explanation

.....

..... [4]

[Total: 11]

Reproduction in Humans P2 questions

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Mark Scheme

Mark schemes will use these abbreviations:

; separates marking points

/ alternatives

() contents of brackets are not required but should be implied

R reject

A accept (for answers correctly cued by the question, or guidance for examiners)

Ig ignore (for incorrect but irrelevant responses)

AW alternative wording (where responses vary more than usual)

AVP alternative valid point (where a greater than usual variety of responses is expected)

ORA or reverse argument

underline actual word underlined must be used by candidate

+ statements on both sides of the + are needed for that mark

O/N18/22/Q2

2(a)(i) (E) urethra ;

(F) bladder ;

(G) kidney ;

(H) ureter ;

R gall (bladder)

2(a)(ii) semen / seminal fluid / sperm / gametes + carried by E / not carried by

H ; 1

2(b)(i) (contains) glucose ; 1

2(b)(ii) insulin gene ; from + human + DNA / chromosome / genome ; To + bacterial + DNA / chromosome / genome / plasmid ; fermenter ; reproduce / multiply / divide / mitosis / binary fission
Ig fermentation

O/N18/22/Q5

5(a) (FSH)

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follicle / egg / ovum + develops / matures / grows ; ovary ; oestrogen + production / release ;
(progesterone) maintains AW uterus + lining / wall ; implantation / pregnancy ; stops LH/FSH +
production / release ;

5(b)(i) fertile ;

sperm in

+ female / vagina / uterus / oviduct / fallopian tube ;sperm survive

+ several days / until ovulation AW / until day 14 ;

5(b)(ii) (name of method)

natural / rhythm / calendar ;

(explanation)

have intercourse outside fertile phase / avoid intercourse in fertile phase ;

no sperm present

+ at same time as egg / at ovulation / on day 14 ;

no fertilisation / no fusion of gametes ;

5(b)(iii) cycles / periods AW

+ irregular / vary in length ;

fertile phase / ovulation

+ on different days ;

data / chart / information + not be available ; sperm may survive for a long time AW

M/J18/21/Q2

2(a)(i) sperm ; **1**

2(a)(ii) testis correctly labelled with **X** ; **1**

2(a)(iii) sperm cells / gamete + meiosis **OR** other cells + mitosis ;

different allele combinations on chromosomes ;

reference to (nucleus of sperm) haploid / half number **AW** ;

reference to restoration of diploid / full number **AW** ;

(upon) fertilisation / fusion of gametes ;

3 Marks can be awarded under *cause of difference or explanation*

2(b) male / sperm + small(er) ;

male / sperm + motile / mobile / can swim **AW** ;

2

2(c) zygote ;

fertilisation ;

Reproduction in Humans P2 questions

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uterus / womb ;

M/J18/21/Q5

5(a)(i) very **AW** small ;
protein coat / capsid ;
no membrane / no cytoplasm / no nucleus / non-cellular / no organelles ;
nucleic acid / DNA / RNA ;
parasitic / (may be) pathogen(ic) / cause disease / non-living **AW** ;
unaffected by antibiotics ;

3

5(a)(ii) reference to immune / immunity (system or affected) ;
reference to reduced **AW** + antibody production ;
reference to reduced **AW** + phagocytic action ;

2

5(b)(i) barrier contraception / correct named contraceptive method e.g. condom ;
abstinence ;
only one partner **AW** ;
reference to use of anti-retroviral drugs ; **2 A** avoid unprotected sex
5(b)(ii) sharing + needles / sharp objects / toothbrushes ;
blood transfusion / contact ;
from mother to baby / across placenta / during birth / during breast-feeding ;

O/N17/22/Q1

1(a) gamete / egg / ovum / sperm correctly named and labelled ;
zygote correctly named and labelled ;

2

1(b) uterus / womb / endometrium ; **1**

1(c) (*child A*) male ;

(*child B*) male ;

(*explanation*)

1 zygote + X + Y ;

2 mitosis ;

3 identical + cells / twins ;

4 A boy / son for both

1(d) **1** yellow body / follicle / corpus luteum / ovary ;

2 placenta ;

3 progesterone ;

4 inhibited / less / no + FSH / LH + production / release ;

5 FSH / LH + pituitary ;

6 FSH + ovum / egg + maturation / development ;

7 LH + ovum / egg + release ;

O/N16/21/Q4(b)

4(b)(i) sterile / not contaminated / no additives ;
antibodies / hormones ;
no need to warm / at correct temperature ;
bonding between mother and child ;
better proportions of nutrients **AW** ;
does not cost money ;
supply / availability on demand ;

4

4(b)(ii) lack of awareness of the benefits ;

Reproduction in Humans P2 questions

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mother unable / chooses not to breast-feed ;
reference to advertising by bottle milk providers ;
not available to do so (e.g. at work / child in care) ;
infection / drug addiction / damaged nipples ;
mother does not produce (enough) milk ;

M/J16/22/Q2

2 (a) C / D / E ;

C / D ;

F ;

E ;

[4]

(b) (i) heroin / alcohol / nicotine / named
addictive drug ;

[1]

(ii) Ig reference to named drug /
substance (e.g. CO, tar)

mother's blood ;

diffusion ;

across or through placenta ;

fetus / baby / embryo + blood ;

umbilical cord / umbilical vein ;

O/N15/21/Q4

4 (a) (i) 11 / 12 day(s) ;

13 / 14 day(s) ;

[2]

A 11th or 12th day

(ii) day 14 ; [1]

(iii) progesterone (concentration) falls / not maintained ;

ref. figure day 24 – 28 ;

progesterone needed to maintain lining / pregnancy ;

[max. 2]

(b) FSH ;

development of follicle / maturation / release of egg / ovum ;

stimulates production of oestrogen ;

oestrogen ;

repairs / builds up / thickens / develops uterus lining ;

maturation of egg / ovum ;

inhibits production of FSH ;

stimulates production of LH ;

LH ;

ovulation / release of egg / ovum ;

[1]

[max. 1]

[1]

[max. 1]

[1]

[1]

[max. 4]

A oestradiol for oestrogen

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Throughout

M/J15/21/Q3

3 (a) amniotic ;
prevents physical harm / damage to
fetus ;

[2]

(b) (i) placenta correctly labelled ;

[1]

(ii) exchange / passage in correct direction
of

+ nutrients / named ;

+ gases / named ;

+ excretory products / named ;

+ antibodies ;

ref. prevent mixing of maternal and
fetal blood ;

(c) male ;

presence of Y (chromosome) ;

R gene

[2]

(d) (i) one extra chromosome / trisomy / three
not two / 47 not 46 ;

position / pair 21 ;

[2]

(ii) Down's syndrome ;

M/J15/22/Q9

9 (a) 1. bacterium / spirochaete /
Treponema ;

2. ulceration / chancre / sore + on
contact part of body ;

3. swollen lymph glands ;

4. rash / description ;

5. flu-like symptoms / examples ;

6. weight loss ;

7. hair loss ;

8. numbness / paralysis / loss of
coordination ;

9. stroke ;

10. dementia ;

11. blindness ;

12. deafness ;

13. heart disease ;

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5. raised temperature / fever /
headache / sore throat

[max 6]

(b) antibiotic / named antibiotic ;

(c) 1. drugs (antiretrovirals / ARVs) can
contain the disease ;

2. stick to one partner / avoid casual
sex ;

3. abstinence ;

4. barrier contraception AW ;

5. screen blood ;

6. don't share / sterilise + needles /
razors / surgical equipment /

toothbrushes ;

7. HIV testing / contact tracing ;

8. education ;

4. Ig contraception unequal

[max 3]

O/N14/21/Q3

3 (a) (i) deeper voice / hair on face or named body part / stronger
muscles / sperm production / larger genitalia ;

[1]

A broadening of shoulders

(ii) testosterone ; [1]

(iii) testes ; [1] A testicles / gonads

(b) (i) F – oestrogen ;

G – progesterone ;

[2]

(ii) ovulation / release of egg or ovum ; [1]

(c) line drawn at 3 weeks $\pm$ 2 squares ;

uterus (lining) increasing in thickness ;

in preparation for receiving (fertilised) ovum / egg ;

ref. time + menstruation ;

[4]

A zygote / embryo

[Total: 10]

M/J14/22/Q4

4 (a)

structure

identified by

letter

name of

structure

carries urine

(yes or no)

carries

sperms (yes

or no)

F ureter yes no

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G urethra yes yes

H rectum no no

J vas

deferens /

sperm duct

no yes

[4]

1 mark per correct row ;

spelling of ureter and urethra must be correct

(b) line drawn across sperm duct ;

line drawn across oviduct ;

[2] R if more than one line drawn on each Fig.– unless

across same structure

R if more than one structure cut

Ig skin cuts

(c) closes / restricts AW the urethra ;

adverse effect on urination AW ;

[2]

Ig ref bladder

Ig refs to pain on urination

0/N13/22/Q9(a)

9 (a) screening blood;

being HIV tested / contact tracing;

avoidance of needle/syringe sharing;

needle exchange schemes / sterilising needles;

condoms / femidoms;

limited partners / no intercourse with prostitutes (who may have multiple partners);

abstinence;

ref. education (about how HIV is spread);

take drugs / follow treatment prescribed (to prevent spread within body); [max 4]

M/J13/21/Q6

6 (a) (i) (FSH) stimulates follicle to develop / egg to develop /

helps to increase numbers of eggs released;

(FSH) stimulates secretion / release of oestrogen;

oestrogen stimulates secretion / release of LH;

LH stimulates ovulation / release of egg;

[max 3]

(ii) (progesterone) prevents / inhibits secretion of FSH;

follicle does not develop / egg unable to mature

/ prevents ovulation A/W;

fertilisation / pregnancy does not occur;

[max 2]

(b) prevent mixing of maternal and fetal blood;

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allow exchange A/W of substances between maternal and fetal blood;
dissolved;
named nutrient;
named gas;
named excretory product;
antibodies / hormones;
[max 5]
Accept nitrogenous waste

M/J12/21/Q7

7 (a) (i) carries urine (R just urea) in male and female;
carries semen / sperms / gametes AW in male; [2]
(ii) ref. size comparison;
ref. qualified numbers comparison;
ref. mobility comparison; [3]
(b) (advantages)
only needs to be undergone once AW (e.g. less trouble);
high reliability / effectiveness AW; [2]
(disadvantages)
does not protect against sexually transmitted diseases;
ref. risks of surgery / anaesthetic;
not a temporary solution;
difficult / expensive to reverse;
need access to medical services AW; [4 max]

O/N11/21/Q7

7 (a) zygote;
division;
mitosis (A anywhere);
blastocyst or described;
implantation AW;
in uterus lining (R wall);
placenta;
membrane(s) or named/amnion/amniotic sac;
named food substance/minerals;
oxygen;
nitrogenous excretion/urea/CO₂;
diffusion;
development of organs/named organs/cells or tissues become specialised; [max 7]
(b) might not be sterile/A ref. possible contamination;
no antibodies;
needs warming/temperature ref.;
less satisfactory bonding;
can lead to obesity in later life AW/wrong proportions of nutrients;
expensive;
supplies may be limited; [max 3]

M/J11/21/Q3

(a) fertilisation;
oviduct / Fallopian tube; [2]
(b) mitosis; [1]

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(c) any 2 genetic similarities

(e.g. sex/eye colour/blood groups/shape of ears/nose etc.)/same genes* once only/

R look alike/same age/same chromosomes;;

characters caused by genes only;

have both inherited identical/same genes* once only;

they both arise from the same zygote/one sperm + one egg; [max 4]

(d) any two environmentally affected characters (e.g. size/weight/hair colour)

R different traits unqualified/intelligence;;

ref. to environment/genes only partly responsible;

named relevant environmental factor e.g. amount of sun light/lack of food etc.; [4]